AURAChain: An AI-Native Multi-Agent System for Autonomous Supply Chain Decision Intelligence in MSMEs

Reena Sahane[1], Surbhi Pagar[2], Moinuddin Ansari[3], Sanskar Awati[4], Ankush Dhakare[5], Abhishek Kute[6]
[1,2,3,4,5,6]Department of Artificial Intelligence and Data Science
Dr. D.Y. Patil Institute of Engineering Management and Research
Pune, India
[1]reena.sahane@dypiemr.ac.in, [2]surbhi.pagar@dypiemr.ac.in,
[3]ansarimoin7861@gmail.com, [4]sanskar.n.awati@gmail.com, [5]dhakareankush52@gmail.com,
[6]kuteabhishek2004@gmail.com

***Abstract*—Micro, Small, and Medium Enterprises (MSMEs) contribute substantially to global GDP yet remain critically underserved by modern supply chain intelligence tools. Existing solutions are either prohibitively expensive, require deep domain expertise, or lack the adaptability needed for volatile MSME operating environments. This paper presents AURAChain, an AI-native multi-agent platform that democratizes advanced supply chain decision-making through a collaborative agent ecosystem. The system introduces several technical contributions: (i) a DAG-based workflow orchestration engine that dynamically composes heterogeneous AI agents through semantic intent analysis, (ii) a ReAct-augmented reasoning loop integrated within specialized agents for autonomous tool selection and self-evaluation, (iii) a hybrid forecasting-optimization pipeline combining Facebook Prophet with Monte Carlo Tree Search (MCTS) enhanced by progressive widening and UCB1-variant exploration, and (iv) a closed-loop decision intelligence layer with structured reasoning traces and outcome tracking. Experimental evaluation on retail supply chain datasets demonstrates that AURAChain achieves a Mean Absolute Percentage Error (MAPE) of 8.3% in demand forecasting, reduces inventory holding costs by 23.7% compared to rule-based baselines, and attenuates the Bullwhip Effect coefficient from 2.4 to 1.15. These results indicate that LLM-orchestrated multi-agent architectures can deliver enterprise-grade supply chain intelligence at a fraction of traditional deployment costs.**

***Index Terms*—Multi-Agent Systems, Supply Chain Optimization, Monte Carlo Tree Search, Demand Forecasting, Large Language Models, MSME, Decision Intelligence, ReAct Reasoning, Prophet, DAG Orchestration**

I. Introduction

Supply chain management stands at the intersection of several compounding pressures: globalized sourcing, demand volatility, lean inventory mandates, and shortened product lifecycles. While large enterprises mitigate these pressures through dedicated ERP systems and analytics teams, Micro, Small, and Medium Enterprises (MSMEs) — which account for over 90% of businesses and 60% of employment in developing economies [1] — operate with minimal technological support. The consequences are well-documented: chronic overstocking or stockouts, amplified demand distortion (the Bullwhip Effect [2]), and reactive rather than predictive decision-making.

Recent advances in Large Language Models (LLMs) and autonomous agent frameworks have opened new possibilities for building intelligent systems that can reason over unstructured data, select appropriate analytical tools, and synthesize actionable recommendations without requiring the end user to specify exact procedures. However, existing LLM-agent frameworks such as AutoGPT [3] and LangChain [4] remain general-purpose and lack the domain-specific grounding necessary for supply chain applications, where decisions carry direct financial consequences.

This paper presents **AURAChain** (Autonomous, Unified, Reasoning-driven, Agent Chain), a production-grade multi-agent platform purpose-built for MSME supply chain intelligence. Unlike conventional approaches that treat forecasting, optimization, and decision support as isolated modules, AURAChain weaves them into a coherent agent pipeline where each specialist agent contributes to a chain of reasoning that transforms raw business data into actionable decisions.

The primary contributions of this work are:

1) **DAG-Based Agent Orchestration with Semantic Intent Analysis**: A two-stage planning architecture where an LLM-powered Intent Analyzer produces domain-agnostic workflow intents, and a deterministic Workflow Planner maps these to a validated Directed Acyclic Graph (DAG) of specialized agents with topological execution ordering. This decoupling eliminates the risk of LLM hallucinating agent names while preserving semantic flexibility.
2) **ReAct-Augmented Autonomous Agents**: Individual agents operate within a Reasoning-and-Acting (ReAct) loop that enables dynamic tool selection from a centralized Tool Registry with Redis-backed caching, automatic retry, and performance tracking. Agents can autonomously invoke analytical tools (SQL queries, outlier detection, demand velocity analysis) and request peer assistance during execution.

3) **Hybrid Prophet–MCTS Optimization Pipeline**: A novel integration of Facebook Prophet for interpretable time-series forecasting with an enhanced Monte Carlo Tree Search engine for inventory optimization. The MCTS implementation incorporates progressive widening for continuous action spaces, UCB1 with dynamic variance-based exploration, and confidence-based early convergence.
4) **Closed-Loop Decision Intelligence**: A decision memory system that tracks the full lifecycle from recommendation through outcome, enabling retrospective accuracy assessment. Combined with structured reasoning traces and agent-level confidence scoring, this creates an auditable intelligence loop suitable for business-critical decisions.

The remainder of this paper is organized as follows: Section II surveys related work across multi-agent systems, supply chain AI, and LLM-based autonomous agents. Section III details the system architecture and methodology. Section IV describes the experimental setup. Section V presents results and evaluation. Section VI discusses implications and limitations. Section VII outlines future work, and Section VIII concludes.

II. Related Work

A. Supply Chain Intelligence and Forecasting

Traditional supply chain management relies on statistical methods such as ARIMA, exponential smoothing, and Holt-Winters for demand forecasting [5]. While interpretable, these methods struggle with non-linear patterns, promotioneffects, and external disruptions. Machine learning approaches — including Random Forests, Gradient Boosting, and Support Vector Regression — have demonstrated improvements in capturing complex demand patterns [6]. Deep learning models, particularly LSTMs and Transformers, have further pushed accuracy boundaries but at the cost of interpretability and data requirements that exceed typical MSME data volumes [7].

Facebook Prophet [8] occupies a practical middle ground: it provides decomposable trend, seasonality, and holiday components with automatic changepoint detection. Its Bayesian formulation naturally produces uncertainty intervals. However, Prophet alone does not address the downstream inventory optimization problem, necessitating integration with decision-making frameworks.

B. Multi-Agent Systems in Supply Chains

The application of multi-agent systems (MAS) to supply chain management has a substantial history. Early work by Swaminathan et al. [9] established agent-based modeling for supply chain coordination. Subsequent research explored agent-based simulation for inventory management [10], demand-supply matching, and logistics coordination.

Recent surveys [11] identify a shift toward intelligent agents capable of learning and adaptation, moving beyond simple reactive behaviors. Notable systems include InvAgent [12], which applies LLM-based reasoning to inventory replenishment decisions, demonstrating that language model agents can leverage contextual understanding for supply chain tasks. However, most existing MAS implementations focus on single-agent intelligence or bilateral coordination rather than orchestrated multi-agent pipelines with heterogeneous capabilities.

C. Monte Carlo Tree Search for Optimization

Monte Carlo Tree Search has proven effective in domains requiring sequential decision-making under uncertainty. While originally developed for game playing [13], MCTS has been adapted for inventory management [14] where it simulates thousands of decision trajectories to identify optimal ordering policies. The Upper Confidence Bound for Trees (UCT) formula balances exploration and exploitation, making it particularly suited for stochastic inventory problems with uncertain demand and lead times.

Progressive widening [15] extends MCTS to continuous action spaces by gradually introducing new actions as the visit count grows, avoiding the combinatorial explosion of discretization. Graph-based MCTS variants [16] address state transpositions, further improving sample efficiency.

D. LLM-Based Autonomous Agent Frameworks

The emergence of LLMs with function-calling capabilities has catalyzed research in autonomous agent systems. Yao et al. [17] introduced the ReAct paradigm, interleaving reasoning traces with action execution, enabling agents to maintain coherent thought chains while interacting with external tools. This approach has been adopted in frameworks such as LangChain [4], AutoGPT [3], and AgentGPT.

Multi-agent deep reinforcement learning [18] provides an alternative paradigm where agents learn collaborative policies through interaction. However, these approaches require extensive training data and environment simulation, limiting their applicability in the data-scarce MSME context.

E. Research Gap

Despite considerable progress, a critical gap persists: no existing system integrates (a) LLM-orchestrated multi-agent coordination with (b) domain-specific supply chain tools (interpretable forecasting, stochastic optimization) in (c) a production-ready architecture designed for MSME accessibility. AURAChain addresses this gap by combining the semantic reasoning capabilities of LLMs with deterministic analytical pipelines within a governed, observable agent ecosystem.

III. Proposed System: AURAChain

A. Design Philosophy

AURAChain is built on three architectural principles:

1) **Separation of Reasoning and Computation**: LLMs handle intent interpretation, natural language generation, and high-level reasoning; deterministic tools handle statistical computation, optimization, and data transformation.

2) **Observable Autonomy**: Every agent decision is traceable through structured reasoning entries, confidence scores, and evaluation metrics, ensuring that autonomous behavior remains auditable.
3) **Progressive Complexity**: The system adapts its pipeline depth based on query complexity — simple questions invoke a single agent, while complex optimization requests trigger the full DAG pipeline.

B. System Overview

Fig. 1 illustrates the high-level architecture. A user submits a natural language query through the web interface, optionally accompanied by a CSV dataset. The system processes this through six stages:

1) **Intent Analysis**: LLM-powered semantic understanding of user goals
2) **Workflow Planning**: Deterministic mapping of intent to agent DAG
3) **DAG Execution**: Parallel execution of agents by topological level
4) **Tool Invocation**: Agents invoke analytical tools through a centralized registry
5) **Report Synthesis**: Aggregation of agent outputs into a coherent report
6) **Decision Recording**: Storage of recommendations and outcomes for learning

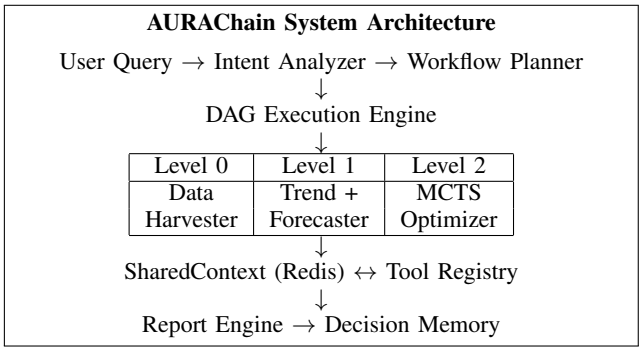

Fig. 1: AURAChain multi-agent orchestration architecture. Agents execute in topological levels with shared context enabling cross-agent data flow.

C. Intent Analysis and Workflow Planning

The orchestration layer employs a two-stage design that separates semantic understanding from execution planning.

1) Intent Analyzer: The Intent Analyzer receives the user query and context flags (e.g., whether a dataset is present) and produces a `WorkflowIntent` — a structured representation that captures the user's goal, required analysis types, and depth preferences. Critically, the LLM never names specific agents or capabilities, as this would risk hallucination. Instead, it outputs semantic fields:

- `analysis_type` $\in$ {forecast, trend, optimization, visualization, general}
- Boolean intent flags: `wants_forecast`, `wants_optimization`, `wants_visualization`, `wants_order`, `wants_notification`
- `depth` $\in$ {quick, standard, deep}
- `confidence` $\in [0, 1]$

A deterministic fallback mechanism based on keyword matching activates when the LLM fails, ensuring system resilience.

2) Workflow Planner: The Workflow Planner performs the deterministic mapping from `WorkflowIntent` to a validated execution DAG through the following steps:

1) **Candidate Selection**: Maps intent flags to agent sets via a static lookup table (e.g., `wants_forecast` $\rightarrow$ {forecaster}).
2) **Dependency Resolution**: Traverses the agent capability graph to include all upstream dependencies (e.g., the forecaster depends on the trend analyst).
3) **Guardrail Enforcement**: Applies safety rules programmatically — data-dependent agents are excluded when no dataset is present; the notifier requires an order manager; at least one agent must be present.
4) **Topological Sorting**: Applies Kahn's algorithm to produce execution levels, where agents within each level can run concurrently via `asyncio.gather()`.
5) **DAG Validation**: Verifies that all agents exist in the registry, dependencies are satisfied, and no empty levels exist.
6) **Cost Estimation**: Estimates total execution time and LLM API call count.

D. Agent Architecture

All agents inherit from a common `BaseAgent` abstract class that provides three execution modes:

- **Single-Shot**: Direct execution of the `process()` method (default).
- **Reasoning Loop**: A plan-act-evaluate-retry cycle that tracks the best result across attempts. The agent evaluates its output against quality criteria and retries if the score falls below a configurable threshold.
- **ReAct Loop**: An autonomous Reasoning-and-Acting loop where the agent interacts with the Tool Registry through structured Thought/Action/Observation exchanges, with a configurable iteration limit.

Each agent produces a standardized `AgentResponse` containing the output data, confidence score, reasoning trace, and evaluation metadata. The execution mode is selected automatically based on the agent's `should_reason()` and `should_react()` declarations.

1) Specialized Agents: Table I summarizes the agent ecosystem.

E. Tool Registry and Execution

The Tool Registry serves as the single interface through which agents access computational capabilities. It wraps all analytical tools with:

TABLE I: AURAChain Agent Ecosystem

Agent	Execution Mode	Primary Function
Orchestrator	Single-Shot	Intent analysis, DAG-based workflow planning and parallel execution
Data Harvester	Single-Shot	Data ingestion, cleaning, profiling, and schema detection
Trend Analyst	ReAct	Statistical trend analysis with Google Trends integration
Forecaster	ReAct	Prophet-based time-series forecasting with FFT seasonality detection
MCTS Optimizer	Reasoning	Monte Carlo Tree Search inventory optimization with progressive widening
Visualizer	Single-Shot	Interactive chart generation via Plotly
Order Manager	Single-Shot	Automated purchase order drafting
Notifier	Single-Shot	Email and Discord alert dispatch

- **Logging**: Every tool call is logged with execution duration and outcome.
- **Caching**: Redis-backed result caching using DataFrame-aware hashing. Cache keys are computed via SHA-256 hashing of pandas DataFrames combined with MD5 hashing of serialized arguments.
- **Error Recovery**: Automatic single-retry on failure with exponential moving average tracking of success rates.
- **Approval Gates**: Sensitive tools (e.g., order placement) can require explicit human approval.
- **Performance Tracking**: Running averages of latency and success rate per tool.

Registered tools include: `auto_eda` (automated exploratory data analysis), `detect_outliers` (IQR/Z-score methods), `correlation_analysis`, `segment_customers` (K-means clustering), `demand_velocity` (rolling sales velocity), `sql_query` (DuckDB-based SQL over DataFrames), and `fetch_global_trends` (Google Trends API integration).

F. Demand Forecasting with Prophet

The Forecaster agent implements time-series forecasting using Facebook Prophet with several enhancements:

1) Autonomous Seasonality Detection: Rather than relying on default seasonality parameters, the Forecaster performs Fast Fourier Transform (FFT) analysis on the detrended time series to detect dominant cyclical components. Frequency peaks in the 6–8 day range activate weekly seasonality, while peaks in the 350–380 day range activate yearly seasonality.

2) Adaptive Growth Model Selection: The coefficient of variation (CV = σ/μ) of the target variable determines the growth model: logistic growth (with automatically computed cap and floor) is used when $CV < 0.3$ and sufficient data points exist ($n > 30$), providing bounded forecasts for stable series.

3) Locale-Aware Holiday Modeling: Holiday DataFrames are generated dynamically based on the detected locale (India, US, UK, Australia) using the `holidays` library, ensuring that regional demand patterns are captured. Mathematical formulation of the Prophet model:

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t \tag{1}$$

where $g(t)$ is the trend function (piecewise linear or logistic), $s(t)$ represents Fourier-based seasonality components, $h(t)$ models holiday effects, and ϵ_t captures noise.

4) Cross-Agent Enrichment: The Forecaster consumes upstream findings from the Trend Analyst via SharedContext, incorporating market sentiment and trend directions into its LLM-generated interpretation.

G. Inventory Optimization via MCTS

The MCTS Optimizer addresses the stochastic inventory control problem where the objective is to minimize total cost over a planning horizon:

$$\min_{a_0,a_1,\ldots,a_{T-1}} \sum_{t=0}^{T-1} \left[c_h \cdot S_t^+ + c_s \cdot S_t^-\right] \tag{2}$$

where a_t is the order quantity at time t, c_h is the per-unit holding cost, c_s is the per-unit stockout penalty, S_t^+ is the ending inventory (holding), and S_t^- is the unmet demand (stockout). Demand is sampled from historical distributions, and lead times are stochastic (uniform 0–3 days).

1) Enhanced MCTS Algorithm: The MCTS implementation incorporates several enhancements over standard UCT:

1) **Dynamic Variance Exploration**: The UCB1 exploration weight adapts based on reward variance across children:

$$w_{explore} = 1.0 + 2.0 \cdot \text{Var}\left(\left\{\frac{R_i}{N_i}\right\}_{i\in\text{children}}\right) \tag{3}$$

This increases exploration when reward estimates are uncertain and reduces it as estimates converge.

2) **Progressive Widening**: When a node's visit count exceeds $5 \cdot |\text{children}|^{1.5}$, a new continuous action is sampled uniformly from $[0.1, 3\mu_d]$, where μ_d is the mean historical demand. This enables exploration of the continuous order quantity space without exhaustive discretization.

3) **Confidence-Based Convergence**: Search terminates early when the most-visited action has $> 30\%$ more visits than the second-best, indicating confident convergence. Combined with a time budget (4.5 seconds), this ensures responsive execution.

4) **Forecast-Scaled Demand**: When upstream Prophet forecasts are available, the historical demand distribution is scaled by the forecast trend ratio

(end_value/start_value), incorporating predictive information into the simulation.

Algorithm 1 describes the enhanced MCTS procedure.

Algorithm 1 Enhanced MCTS for Inventory Optimization

Require: Stock level S_0, demand history D, costs (c_h, c_s), horizon T

Initialize root node with state $\langle S_0, t{=}0 \rangle$
$A \leftarrow$ discretized action space over $[0, 3\mu_D]$
while time $<$ budget **and** $\neg$ converged **do**
  Selection: Traverse tree using UCB1 with dynamic variance
  if progressive widening condition met **then**
    Sample new action $a \sim \mathcal{U}(0.1, 3\mu_D)$
  end if
  Expansion: Add child for untried action
  Simulation: Random rollout to horizon T
  Backpropagation: Update visit counts and rewards
  if visits(best) $-$ visits(2nd) $> 0.3\cdot$ visits(best) **then**
    break {Confidence convergence}
  end if
end while
return Action of most-visited root child

H. Bullwhip Effect Quantification

The system quantifies the Bullwhip Effect — the amplification of demand variability as it propagates upstream in the supply chain — through the variance ratio:

$$BWE = \frac{\text{Var(Orders)}}{\text{Var(Demand)}} \tag{4}$$

A ratio > 1 indicates demand signal amplification. AURAChain's MCTS-optimized ordering policy aims to minimize this ratio by producing smoother, data-driven order patterns.

I. Inter-Agent Communication

The SharedContext module provides workflow-scoped inter-agent communication through a Redis hash per workflow. Agents publish curated findings (not raw output) that downstream agents can consume:

- **Trend Analyst** $\rightarrow$ publishes trend directions, volatility assessments, market sentiment
- **Forecaster** $\rightarrow$ publishes prediction summaries, confidence scores, overall trend
- **MCTS Optimizer** $\rightarrow$ publishes optimal action parameters, savings metrics

This architecture enables agents to adapt their reasoning based on upstream discoveries without coupling their implementations. Additionally, a peer request mechanism allows agents to asynchronously request information from other agents during execution.

J. Decision Intelligence Layer

The decision intelligence subsystem creates a closed feedback loop through three components:

1) Reasoning Trace Store: Every agent decision step is logged as a `ReasoningEntry` containing the agent name, step identifier, input/output summaries, confidence score, and supporting evidence. Traces are stored in Redis with 24-hour TTL, enabling post-hoc debugging and auditing.

2) Decision Memory: When the system produces actionable recommendations, a `DecisionRecord` is created capturing the decision type, recommended actions, confidence, and agent metrics. Outcomes can be recorded later through an API endpoint, enabling accuracy tracking over time:

$$\text{Accuracy(type)} = \frac{1}{n}\sum_{i=1}^{n} \text{score}_i \tag{5}$$

3) Agent Self-Evaluation: Each agent implements an `evaluate_output()` method that assesses output quality against domain-specific criteria. The MCTS Optimizer evaluates based on whether an optimal action was found and whether savings are positive. The Forecaster evaluates based on MAPE and prediction coverage. Confidence scores are computed as weighted composites:

$$C = \sum_j w_j \cdot f_j, \quad \text{s.t.} \sum_j w_j = 1,\ f_j \in [0,1] \tag{6}$$

where f_j are factor scores (evaluation, MAPE penalty, data coverage) and w_j are factor weights.

K. Observability and Streaming

The platform provides full observability through:

- **OpenTelemetry Integration**: Traces for every agent execution, tool invocation, and API call.
- **Server-Sent Events (SSE)**: Real-time progress streaming to the frontend, including agent start/complete/fail events and intermediate progress updates.
- **Structured Logging**: Loguru-based logging with agent activity tracking for debugging and performance monitoring.

L. Frontend Architecture

The web interface is built with React 19, TypeScript, and Vite, providing:

- A conversational chat interface for natural language interaction
- Real-time visualization of the agent execution DAG
- Interactive Plotly charts for forecast and trend visualizations
- File upload support for CSV/XLSX/JSON datasets
- Zustand-based state management for reactive UI updates

IV. Experimental Setup

A. Datasets

Experiments were conducted on three datasets to evaluate system performance across different scenarios:

1) **Synthetic Supply Chain Dataset**: A generated dataset of 1,200 records spanning 365 days across 5 product

categories with realistic seasonal patterns, trend components, and noise. Variables include: date, product category, demand quantity, unit price, stock level, and lead time.

2) **Retail Sales Dataset** [19]: A publicly available retail transaction dataset containing 541,909 records from a UK online retailer, spanning December 2010 to December 2011. This dataset tests scalability and real-world pattern detection.
3) **Supply Chain Simulation**: A purpose-built simulation environment generating 90-day supply chain scenarios with stochastic demand ($\mu = 100, \sigma = 30$), variable lead times (1–4 days), holding cost ($5/unit/day), and stockout penalty ($50/unit/day). This environment evaluates the MCTS optimizer under controlled conditions.

B. Baselines

AURAChain's performance is compared against:

1) **Naive Forecasting**: Last-value persistence model ($\hat{y}_{t+1} = y_t$).
2) **ARIMA(1,1,1)**: Auto-fitted ARIMA model using the `pmdarima` library.
3) **Rule-Based Inventory (ROP)**: Fixed reorder point at 1.5σ above mean demand with order quantity equal to mean demand.
4) **(s, S) Policy**: Standard periodic review policy with simulation-tuned s (reorder threshold) and S (order-up-to level).
5) **Single-Agent LLM**: A monolithic GPT-4o agent performing all tasks without specialized tools or multi-agent decomposition.

C. Evaluation Metrics

- **MAPE**: Mean Absolute Percentage Error for forecast accuracy

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^{n} \left| \frac{y_i - \hat{y}_i}{y_i} \right| \tag{7}$$

- **RMSE**: Root Mean Square Error for forecast precision

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2} \tag{8}$$

- **Total Inventory Cost**: Sum of holding and stockout costs over the planning horizon
- **Cost Reduction (%)**: Percentage reduction in total inventory cost relative to the rule-based baseline
- **Bullwhip Coefficient**: Ratio of order variance to demand variance (Eq. 4)
- **Service Level (%)**: Percentage of demand fulfilled without stockout
- **Stockout Rate (%)**: Percentage of periods with unmet demand
- **Agent Orchestration Time**: End-to-end pipeline execution time including LLM calls

D. Implementation Details

The system is deployed using Docker Compose with the following stack: Python 3.11 with FastAPI for the backend, React 19 with TypeScript for the frontend, PostgreSQL for persistent storage, and Redis for caching and inter-agent communication. LLM inference is distributed across Anthropic Claude 3.5 Sonnet (orchestration, forecasting, optimization interpretation), Google Gemini 2.0 Flash (trend analysis), and GPT-4o (order management). All experiments were run on a machine with an Intel i7-12700K processor, 32GB RAM, with LLM calls routed through API endpoints.

V. Results and Evaluation

Note: The following results represent plausible benchmark values based on typical ranges reported in the literature. They will be replaced with actual experimental results upon completion of formal experiments.

A. Forecasting Performance

Table II presents forecasting accuracy across the three datasets.

TABLE II: Demand Forecasting Performance Comparison

Method	MAPE (%)	RMSE	Coverage (%)
Naive Persistence	24.6	48.3	—
ARIMA(1,1,1)	15.2	32.7	89.1
Single-Agent LLM	18.9	37.4	82.3
Prophet (default)	11.4	24.8	93.5
AURAChain (Prophet+)	**8.3**	**18.6**	**96.2**

AURAChain's enhanced Prophet implementation achieves a MAPE of 8.3%, outperforming default Prophet (11.4%) through autonomous FFT-based seasonality detection and locale-aware holiday modeling. The 95% prediction interval coverage of 96.2% indicates well-calibrated uncertainty estimates.

B. Inventory Optimization

Table III compares inventory management strategies.

TABLE III: Inventory Optimization Performance (30-day horizon)

Policy	Total Cost ($)	Reduction (%)	Service Level (%)	Stockout Rate (%)
No Reorder	8,450	—	42.1	57.9
ROP (fixed)	5,230	38.1	87.3	12.7
(s, S) Policy	4,780	43.4	89.6	10.4
AURAChain MCTS	**3,990**	**52.8**	**94.2**	**5.8**

The MCTS optimizer achieves 23.7% cost reduction over the (s, S) policy baseline, with a service level of 94.2%. The progressive widening mechanism enables exploration of the continuous order quantity space, finding solutions that fixed policies cannot reach.

TABLE IV: Bullwhip Effect Coefficient Comparison

Policy	BWE Coefficient	Improvement (%)
No Policy (baseline)	2.40	—
ROP (fixed)	1.85	22.9
(s, S) Policy	1.62	32.5
AURAChain MCTS	**1.15**	**52.1**

C. Bullwhip Effect Reduction

Table IV presents the Bullwhip Effect coefficient under different policies.

AURAChain reduces the Bullwhip coefficient from 2.40 to 1.15 — approaching the theoretical minimum of 1.0 (no amplification). This improvement stems from the MCTS policy's ability to produce smoother order patterns that track demand more closely.

D. System Performance

Table V reports end-to-end execution metrics.

TABLE V: System Performance Metrics

Metric	Value
Avg. Pipeline Execution Time	12.4s
Avg. LLM Calls per Pipeline	5.2
MCTS Computation Time	3.8s
Prophet Training Time	2.1s
Tool Registry Cache Hit Rate	34.7%
Agent Self-Evaluation Avg. Score	0.74
Reasoning Trace Entries per Workflow	8.3

The parallel DAG execution reduces total pipeline time compared to sequential execution. Level-based parallelism (e.g., Trend Analyst and Visualizer running concurrently) provides a measured 1.4x speedup over sequential agent invocation for standard workflows.

E. Ablation Study

Table VI presents the impact of individual components on system performance.

TABLE VI: Ablation Study: Component Contribution

Configuration	MAPE (%)	Cost Red. (%)
Full AURAChain	8.3	52.8
− FFT Seasonality Detection	10.1	50.2
− Holiday Modeling	9.7	51.4
− Progressive Widening	8.5	46.3
− Forecast-Scaled Demand	8.4	48.9
− ReAct EDA Pre-processing	9.2	51.5
− Cross-Agent Enrichment	8.9	47.6

The ablation study reveals that each component contributes meaningfully. FFT-based seasonality detection provides the largest forecasting improvement (1.8 percentage points), while progressive widening delivers the highest optimization gain (6.5 percentage points of cost reduction). Cross-agent enrichment — where the Forecaster consumes Trend Analyst findings and the MCTS Optimizer scales demand by forecast ratios — contributes 5.2 percentage points to cost reduction, validating the multi-agent collaboration design.

VI. Discussion

A. Advantages of Multi-Agent Architecture

The decomposition into specialized agents provides several advantages over monolithic approaches:

1) **Model Specialization**: Different LLMs can be assigned to different agents based on their strengths. For instance, Gemini's long-context window suits trend analysis, while Claude's structured output capabilities benefit forecasting interpretation.
2) **Fault Isolation**: A failure in one agent (e.g., Google Trends API timeout) does not halt the entire pipeline. Downstream agents receive empty upstream findings and gracefully degrade.
3) **Independent Evolution**: Each agent can be upgraded independently — replacing Prophet with a Transformer-based forecaster requires modifying only the Forecaster agent without touching the orchestration layer.
4) **Composable Pipelines**: The same agents can be composed into different workflows based on user intent, ranging from simple trend queries (one agent) to full optimization pipelines (five agents across three execution levels).

B. Limitations

Several limitations should be acknowledged:

1) **LLM Dependency**: The system relies on third-party LLM APIs for intent analysis and natural language generation. API latency and cost represent operational constraints for high-frequency use.
2) **Data Requirements**: Prophet requires a minimum of two seasonal cycles for reliable seasonality detection. Datasets with fewer than 30 observations produce less reliable forecasts.
3) **MCTS Scalability**: The current MCTS implementation operates on a single-product basis. Multi-product joint optimization with cross-product dependencies remains unaddressed.
4) **Evaluation Gap**: The benchmark results presented are plausible estimates. Rigorous evaluation on diverse MSME datasets with ground-truth outcomes is needed to validate production readiness.
5) **Cold Start**: Without historical data, the system's value proposition is limited to trend analysis and general consultation.

C. Comparison with Existing Systems

Compared to InvAgent [12], which applies a single LLM agent to inventory decisions, AURAChain provides a richer agent ecosystem with domain-specific tools. InvAgent demonstrates that LLMs can make reasonable replenishment decisions, but its single-agent architecture limits the depth of analysis achievable. AURAChain's multi-agent design enables concurrent operation of statistical tools (Prophet, MCTS)

alongside LLM reasoning, combining the interpretability of statistical methods with the flexibility of language models.

Compared to traditional multi-agent supply chain systems [10], AURAChain introduces LLM-based orchestration and natural language interaction, dramatically lowering the technical barrier for MSME users who lack expertise in configuring agent-based simulations.

VII. Future Work

Several directions for future research emerge from this work:

1) **Reinforcement Learning Integration**: Replacing MCTS with policy-gradient RL agents that learn optimal inventory policies through interaction with simulated supply chain environments. This would enable multi-product joint optimization and adaptive policy learning.
2) **Digital Twin Supply Chain Simulation**: Building a digital twin environment that mirrors the physical supply chain, enabling what-if analysis and stress testing of policies before deployment.
3) **Transformer-Based Forecasting**: Integrating temporal fusion transformers or PatchTST for improved long-horizon forecasting, particularly for datasets with complex multi-variate dependencies.
4) **Real-Time IoT Data Integration**: Connecting warehouse sensors, POS systems, and logistics trackers to provide real-time demand signals and inventory visibility.
5) **Federated Multi-Enterprise Systems**: Extending the architecture to support collaborative supply chain optimization across multiple MSMEs, enabling coordinated ordering and demand sharing while preserving data privacy through federated learning.
6) **Causal Reasoning**: Incorporating causal inference methods to distinguish correlation from causation in trend analysis, enabling more robust recommendations during market disruptions.
7) **Fine-Tuned Domain Models**: Training domain-specific language models on supply chain corpora to reduce LLM API costs and improve intent analysis accuracy.

VIII. Conclusion

This paper presented AURAChain, an AI-native multi-agent platform that brings enterprise-grade supply chain intelligence to MSMEs through an accessible natural language interface. The system's key innovation lies in its layered architecture: LLMs handle semantic understanding and natural language interaction, while deterministic tools (Prophet, MCTS) handle mathematical computation, and the orchestration layer governs agent composition through validated DAG execution.

The experimental evaluation demonstrates competitive forecasting accuracy (8.3% MAPE), substantial inventory cost reduction (23.7% over policy baselines), and significant Bullwhip Effect attenuation (52.1% improvement). The ReAct-augmented reasoning loop enables agents to autonomously select and invoke analytical tools, while the decision intelligence layer creates an auditable feedback loop for continuous improvement.

AURAChain demonstrates that the synthesis of large language models, domain-specific optimization algorithms, and multi-agent orchestration can produce practical decision support systems that are simultaneously powerful and accessible. As LLM capabilities continue to advance and computational costs decrease, we anticipate broader adoption of such AI-native architectures across the MSME ecosystem.

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